MIL-S-3534D
11 September 1974
SUPERSEDING
MIL-S-3534C
9 November 1967

MILITARY SPECIFICATION

SURVEYING INSTRUMENTS AND ACCESSORIES,

PACKAGING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for preservation, packaging, packing, and marking of surveying instruments and accessories.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

NNN-P-40	- Paper, Lens.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

Military

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.

FSC PACK

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STANDARDS

Federal

FED. TEST METHOD STD. No. 101	- Preservation, Packaging and Packing Materials: Test Procedures.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC, AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section,1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination and test within the time frame specified (see 6.2) to prove,

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prior to starting production packaging and packing, that the applied preservation, packaging, packing and marking comply with the requirements of this specification. Examination and tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be accomplished utilizing either the preproduction model of the equipment or a production unit. If the preproduction model of the equipment is utilized, any preservation, packaging and packing shall be removed by the supplier at no expense to the Government, when requested by the Government to facilitate comparison between the preproduction model of the equipment and production units.

3.2 Materials. Materials shall be as specified herein and in applicable specifications. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification. Materials shall be free of defects affecting serviceability.

3.3 Preservation and packaging. Preservation and packaging shall be Level A, B, or C as specified (see 6.2).

3.3.1 Level A.

3.3.1.1 Cleaning and drying. All metal surfaces of the instruments shall be cleaned and dried in accordance with MIL-P-116, Cleaning Process C-1 and Drying Process D-4. Lens and mirror surfaces shall be cleaned with tissue paper conforming to NNN-P-40, Type I.

3.3.1.2 Methods of preservation. Methods of preservation indicated by symbols herein shall correspond to the applicable symbols listed in MIL-P-116.

3.3.1.3 Preservative and application. Immediately after cleaning and drying, exposed threads and other unpainted or unprotected surfaces of instruments and accessories, except those instruments and accessories specified to be preserved Method II herein, requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with a thin film of Type P-7 preservative conforming to the applicable specification listed in MIL-P-116. The preservative shall be applied by brushing or wiping. Care shall be exercised to prevent the preservative from contacting lenses, mirrors, instrument finishes, or nonmetallic surfaces. Surfaces to which the preservative is applied shall be wrapped or covered with barrier material conforming to MIL-B-121, Type II, Grade A, Class 2, and the wrap or cover shall be secured with tape conforming to PPP-T-76. The tape shall be applied on the barrier material and not in contact with the instrument finish.

3.3.1.4 Alidades.

3.3.1.4.1 Surveyor, triangular, boxwood. Each alidade with technical publications shall be individually preserved, Method IC-1.

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3.3.1.4.2 Surveyor, telescope and surveyor, open sight. Each instrument and its accessories, tools, and publications contained in a carrying case and each instrument and its accessories, tools, and publications, not contained in a carrying case shall be preserved Method IIb. The inner container shall be the carrying case when furnished or a close-fitting box, Domestic Class, SW, Grade 125, when a case is not furnished and the outer container shall be W5c or W6c, as applicable. Both containers shall conform to PPP-B-636, style optional. The outer container shall be waterproofed sealed with tape in accordance with the appendix to the box specification, Method V closure.

3.3.1.5 Astrolabes. Each astrolabe, accessories, tools, and publications shall be preserved Method IIb. The carrying case for the astrolabe shall be the inner container. The outer container shall be a close-fitting box conforming to PPP-B-636, W5c, W6c, or V3c, style optional, as applicable to the weight of the astrolabe. The outer container shall be waterproof sealed with tape in accordance with the appendix to the box specification, Method V closure.

3.3.1.6 Levels, astronomical attachments, heliotropes, collimators and transits, with or without carrying cases. Each instrument and its accessories, tools, and publications shall be preserved Method IIb. The inner container shall be the carrying case, when furnished, or a close-fitting box, Class Domestic SW, Grade 125, when a case is not furnished. The outer container shall be a close-fitting box, W5c, W6c, or V3c, as applicable to the weight of the item. Both containers shall conform to PPP-B-636, style optional. The outer container shall be waterproof sealed with tape in accordance with the appendix to the box specification, Method V closure.

3.3.1.7 Altimeters. Each altimeter shall be preserved Method IIb. The altimeter case containing the desiccant shall be considered the inner container. Technical publications and the altimeter case shall be enclosed in the sealed Method II barrier. The sealed barrier shall be overwrapped with a 2 inch thickness of cushioning conforming to PPP-C-843, Type I, Class A, secured in place with tape conforming to PPP-T-76. The cushioned altimeter shall be placed in a box conforming to PPP-B-636, W5c, style optional. The box shall be waterproof sealed with tape in accordance with the appendix to the box specification, Method V closure.

3.3.1.8 Clinometers, hand levels, rod levels and striding levels. Each instrument, accessories, and technical publications contained in a carrying case or in a vendor's carton shall be preserved Method IA-8.

3.3.1.9 Planetable board. Each planetable board with technical publications shall be preserved Method IA-8.

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3.3.1.10 Theodolites.

3.3.1.10.1 Hermetically sealed. The rigid metal hermetically sealed case with the desiccant inside shall not require additional preservation, except that when a transport case is furnished, the theodolite shall be placed in the transport case.

3.3.1.10.2 Not hermetically sealed. Each case containing the theodolite, tools, accessories, and technical publications shall be preserved Method IIb, using the case as the inner container and a close-fitting box conforming to PPP-B-636, V3c, style optional, as the outer container. The box shall be waterproof sealed with tape in accordance with the appendix to the box specification, Method V closure.

3.3.1.10.3 Battery and lighting accessory case. Each battery and lighting accessory case with all accessories contained within the case shall be preserved Method III. The container shall be a box conforming to PPP-B-636, W5c, W6c or V3c, as applicable to the weight of the case, style optional. The box shall be waterproof sealed with tape in accordance with the appendix to the box specification, Method V closure.

3.3.1.10.4 Tool and accessory case. Each tool and accessory case containing the tools and accessories shall be preserved Method IC-1.

3.3.1.10.5 Winterization kit. Each winterization kit shall be preserved Method IC-1.

3.3.1.10.6 Knapsack. Each knapsack shall be preserved Method III. The container shall be a box conforming to PPP-B-636, W5c, style optional. The box shall be waterproof sealed with tape in accordance with the appendix to the box specification, Method V closure.

3.3.1.11 Tribach with optical plummet and base plate. Each tribach with optical plummet and base plate and technical publications shall be preserved Method IA-14. The inner container shall be a close-fitting box, Class Domestic, SW, Grade 125 and the outer container shall be a close-fitting box, W5c, W6c or V3c, as applicable to the weight of the item, style optional. Both containers shall conform to PPP-B-636. The outer container shall be waterproof sealed with tape in accordance with the appendix to the box specification, Method V closure.

3.3.1.12 Level rods, range poles, stadia rods and tripods, with or without carrying cases. Each rod, pole or tripod shall be unjointed, folded, retracted or collapsed to the minimum dimensions, but not to the extent they will not fit the carrying case when provided. Each rod, pole, or tripod with or without carrying case shall be preserved Method III. Containers shall be weather-resistant type, compliance symbol as applicable, style optional, conforming to PPP-B-636. For items furnished without cases the ends of the containers next to the shoes of range poles and tripods shall be fitted with a wood block to prevent damage to the container from the shoes. The containers shall be waterproof sealed with tape in accordance with the appendix to the box specification, Method V closure.

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3.3.1.13 Trivets. Each trivet shall be preserved Method IC-1.

3.3.1.14 Arrow sets. Each set of arrows on a ring shall be preserved Method III. The containers shall be boxes conforming to PPP-B-636, W5c, style optional. The boxes shall be waterproof sealed with tape in accordance with the appendix to the box specification, Method V closure.

3.3.1.15 Plumb bobs. Each plumb bob with or without a sheath shall be preserved Method IC-1. Plumb bobs without sheaths shall be placed in a vendor's carton prior to placing in the bag.

3.3.1.16 Other instruments. Other instruments not definitely specified herein shall be preserved in accordance with the method(s) criteria of MIL-P-116, determined as designated under the intended use of MIL-P-116.

3.3.1.17 Repair parts, tools and publications. Repair parts, tools and publications for each instrument, if not definitely specified herein, shall be packaged with the instrument.

3.3.1.18 Intermediate packaging. Items of like description preserved Method IC-1 or Method IA-8 shall be intermediate packaged together in quantities not to exceed the weight limitation of the boxes, in boxes conforming to PPP-B-636, weather-resistant class, style optional. The boxes shall be waterproof sealed in accordance with the appendix to the box specification, Method V closure.

3.3.2 Level B. Unless otherwise specified (see 6.2 and 6.3), Level B preservation and packaging requirements shall be as follows: The component items shall be preserved as specified for Level A, except items preserved Method IA-8 shall be preserved Method IC-1, and intermediate containers shall be Domestic Class, SW, grade as required, style optional. Waterproof sealing with tape shall not be required.

3.3.3 Level C. The instruments, accessories, tools and publications shall be preserved and packaged to afford protection against deterioration and damage during shipment and handling from the supplier to the initial destination. The supplier's standard practice will be acceptable when if fulfills these requirements.

3.4 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

3.4.1 Level A.

3.4.1.1 Theodolites. Each theodolite, with accessories as specified shall be packed in a close-fitting box conforming to PPP-B-601, Overseas Type, Style

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A or B. Accessories to be packed with the theodolite that are preserved Method IC-1 shall be consolidated packaged in a close-fitting box conforming to PPP-B-636, W6c, style optional, before packing is accomplished. The components shall be blocked braced and cushioned within the box in accordance with MIL-STD-1186. The theodolite shall be completely isolated from the box sides, ends, top and bottom and the packed accessories with cushioning conforming to PPP-C-843, Type I, Class A. The box lid shall be secured with wood screws. Strapping shall be zinc coated (see 6.5).

3.4.1.2 Instruments or accessories, excluding theodolites. Instruments or accessories of like description, including those intermediate packaged shall be packed together in boxes specified in 3.4.1.1. Boxes containing instruments or accessories preserved Method II shall have the lids secured with wood screws. Strapping shall be zinc coated (see 6.5). Items preserved and packaged as specified in 3.3 shall be placed and secured in the transport case prior to packing, when a transport case is furnished and the transport case design will permit securing the preserved item within the case.

3.4.2 Level B. Theodolites, instruments or accessories shall be packed as specified in 3.4.1.1 or 3.4.1.2 except that boxes shall conform to PPP-B-601, Domestic Type, style optional or PPP-B-640, Class 2, style optional. Boxes conforming to PPP-B-640 shall be reinforced with bands of reinforced filament tape applied in accordance with the appendix to the box specification. Strapping will not be required to be zinc coated.

3.4.3 Level C. Theodolites, instruments or accessories shall be packed to assure carrier acceptance and safe delivery to destination at lowest rating in containers complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

3.5 Marking. Marking shall be in accordance with MIL-STD-129.

3.5.1 Special marking. Theodolites packed Level A or B shall have the exterior of the container marked with an arrow to indicate up.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Material inspection. The supplier is responsible for insuring that supplies and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction pack inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction pack inspection.

4.3.1 Examination. The preproduction pack shall be examined as specified in 4.4.4. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The preproduction pack of the boxed components of Level A or B shall be tested in accordance with 4.3.2.1. Shifting of contents, any visible damage to the contents, deformation of containers within the pack, or shifting of packages within the pack shall be cause for rejection.

4.3.2.1 Boxed equipment (Level A or B). Boxed equipment not exceeding 200 pounds shall be subjected to the free-fall drop test in accordance with Test Method 5007 of FED. TEST METHOD STD. No. 101. Boxed equipment exceeding 200 pounds shall be subjected to the pendulum impact test in accordance with Test Method 5012 of FED. TEST METHOD STD. No. 101.

4.4 Quality conformance inspection.

4.4.1 Inspection stages. 1/ Inspection shall be performed in two stages as follows:

(a) The first stage shall include inspection of the methods and materials used during the cleaning, drying, and preservation of instruments packaged in accordance with Method II. Inspection shall be performed prior to application of the Method II barrier.
(b) The second stage shall include inspection of methods and materials used in packaging and packing of instruments subsequent to application of the Method II barrier.

1/ Items not preserved Method II shall be inspected for all appropriate first and second stage defects at the time they are offered for shipment. The unit of product shall be as described in 4.4.2(b).

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4.4.2 Unit of product. For purposes of inspection, the unit of product shall be as follows:

(a) First stage. A surveying instrument cleaned and preserved but prior to application of the Method II barrier.
(b) Second stage. A surveying instrument or an item(s) of associated equipment prepared for shipment.

4.4.3 Sampling. Sampling for first and second stage examination shall be in accordance with MIL-STD-105.

4.4.4 Examination. Samples selected in accordance with 4.4.3 shall be examined for the defects of the appropriate stage marked "X" for the applicable level in Table I. AQL shall be 2.5 percent defective for each inspection stage.

Table I. Classification of Defects

	Defect	Level A	B	C
	Both Stages			
101.	Materials not as specified for preservation, packaging and packing. Each incorrect material shall constitute one defect (see 3.2).	X	X	-
	1st Stage			
102.	Surfaces not cleaned and dried as specified (see 3.3.1.1, 3.3.2).	X	X	-
103.	Preservation methods not as specified. Each incorrect method shall constitute one defect (see 3.3.1.3, 3.3.2).	X	X	-
104.	Application of preservatives not as specified (see 3.3.1.3, 3.3.2).	X	X	-
105.	Preservative in contact with lenses, mirrors or instrument finish (see 3.3.1.3, 3.3.2).	X	X	-
106.	Tape applied to instrument finish (see 3.3.1.3, 3.3.2).	X	X	-
	2nd Stage			
107.	Method II barrier not applied as specified (see 3.3.1.4.2, 3.3.1.5, 3.3.1.6, 3.3.1.7, 3.3.1.10.2, 3.3.2).	X	X	-
108.	Outer container for Method IIb package not waterproof sealed with tape (see 3.3.1.4.2, 3.3.1.5, 3.3.1.6, 3.3.1.7, 3.3.1.10.2, 3.3.2).	X	X	-
109.	Containers for Method III items not waterproof sealed with tape (see 3.3.1.10.3, 3.3.1.10.6, 3.3.1.12, 3.3.1.14, 3.3.2).	X	X	-

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Table I. Classification of Defects (Cont'd)

Defect	Level A	Level B	Level C
2nd Stage (Cont'd)			
110. Outer container for Method IA-14 not waterproof sealed with tape (see 3.3.1.11).	X	X	-
111. Repair parts, tools and publications not packaged with the instrument for which intended (see 3.3.1.17, 3.3.2).	X	X	-
112. Intermediate packaging not as specified (see 3.3.1.18, 3.3.2).	X	X	-
113. Theodolites not packed as specified (see 3.4.1.1, 3.4.2).	X	X	-
114. Theodolite not cushioned as specified (see 3.4.1.1, 3.4.2).	X	X	-
115. Strapping not zinc coated (see 3.4.1.1, 3.4.1.2).	X	-	-
116. Marking illegible, incomplete, incorrect or missing (see 3.5).	X	X	X

5. PREPARATION FOR DELIVERY (This section not applicable to this specification.)

6. NOTES

6.1 Intended use. It is intended that this specification be used for preservation, packaging, packing and marking of surveying instruments and accessories and be used for reference in Section 5 of commodity specifications or for direct reference in contract or orders. It is also intended to be used in like manner for other instruments and accessories similar in basic features and design although not definitely specified by nomenclature in this specification.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction pack (see 3.1).
(c) Level of preservation and packaging and level of packing required (see 3.3 and 3.4). Level B preservation and packaging (see 3.3.2 and 6.3) is intended to provide limited **but** economical protection and should be specified only when it is determined the surveying instruments and accessories will be held in covered storage for an indefinite period.

6.3 Level B. Preservation and packaging (see 3.3.2) has been developed for use by the Army and Air Force for specific conditions specified in 6.2. For Navy procurements, details of Level B preservation and packaging, when required, should be developed at time of procurement when all conditions are known.

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6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack and mark the surveying instruments and accessories in accordance with this specification.

6.5 Army policy permits the use of Finish A (organic coated) strapping for Army-sponsored shipments.

Custodians:

Army - ME
Navy - YD
Air Force - 32

Review activity:

Army - SM

Preparing activity:

Army - ME

Project No. PACK-0365

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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